US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250281191
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	BETELIA; Rainier et al.

SHOCK WAVE CATHETER WITH RETRACTABLE ENCLOSURE

Abstract

A shock wave catheter for treating a lesion in a body lumen includes a shock wave emitter assembly and a retractable enclosure. In a first configuration of the catheter, the retractable enclosure is sealed to a distal member of the catheter and the lesion is treated by shock waves propagating through the enclosure wall. In a second configuration of the catheter, the retractable enclosure is retracted from the distal member of the catheter and the lesion is treated directly by shock waves emitted from the shock wave emitter assembly.

Inventors: **BETELIA; Rainier (San Jose, CA), JENKINS; Todd Weston (San Jose, CA), ROND; Danielle Sheree (Santa Clara, CA)**

Applicant: **Shockwave Medical, Inc.** (Santa Clara, CA)

Family ID: **94255997**

Assignee: **Shockwave Medical, Inc. (Santa Clara, CA)**

Appl. No.: **18/596388**

Filed: **March 05, 2024**

Publication Classification

Int. Cl.: **A61B17/22** (20060101); **A61B17/00** (20060101); **A61M25/10** (20130101)

U.S. Cl.:

CPC **A61B17/22022** (20130101); A61B2017/00336 (20130101); A61B2017/22008 (20130101); A61B2017/22025 (20130101); A61B2017/22061 (20130101); A61M25/104 (20130101)

Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates generally to the field of medical devices and methods, and more specifically to shock wave catheter devices for treating calcified lesions in body lumens, such as calcified lesions and occlusions in vasculature and kidney stones in the urinary system.

BACKGROUND

[0002] A wide variety of catheters have been developed for treating calcified lesions, such as calcified lesions in vasculature associated with arterial disease. For example, treatment systems for percutaneous coronary angioplasty or peripheral angioplasty use angioplasty balloons to dilate a calcified lesion and restore normal blood flow in a vessel. In these types of procedures, a catheter carrying a balloon is advanced into the vasculature along a guide wire until the balloon is aligned with calcified plaques. The balloon is then pressurized (normally to greater than 10 atm), causing the balloon to expand in a vessel to push calcified plaques back into the vessel wall and dilate occluded regions of vasculature.

[0003] More recently, the technique and treatment of intravascular lithotripsy (IVL) has been developed, which is an interventional procedure to modify calcified plaque in diseased arteries. The mechanism of plaque modification is through use of a catheter having one or more acoustic shock wave generating sources located within a liquid that can generate acoustic shock waves that modify the calcified plaque. IVL devices vary in design with respect to the energy source used to generate the acoustic shock waves, with two exemplary energy sources being electrohydraulic generation and laser generation.

[0004] For electrohydraulic generation of acoustic shock waves, a conductive solution (e.g., saline) may be contained within an enclosure that surrounds electrodes or can be flushed through a tube that surrounds the electrodes. The calcified plaque modification is achieved by creating acoustic shock waves within the catheter by an electrical discharge across the electrodes. This discharge creates one or more rapidly expanding vapor bubbles that generate the acoustic shock waves. These shock waves propagate radially outward and modify calcified plaque within the blood vessels. For laser generation of acoustic shock waves, a laser pulse is transmitted into and absorbed by a fluid within the catheter. This absorption process rapidly heats and vaporizes the fluid, thereby generating the rapidly expanding vapor bubble, as well as the acoustic shock waves that propagate outward and modify the calcified plaque. The acoustic shock wave intensity is higher if a fluid is chosen that exhibits strong absorption at the laser wavelength that is employed. These examples of IVL devices are not intended to be a comprehensive list of potential energy sources to create IVL shock waves.

[0005] The IVL process may be considered different from standard atherectomy procedures in that it cracks calcium but does not liberate the cracked calcium from the tissue. Hence, generally speaking, IVL should not require aspiration or embolic protection. Further, due to the compliance of a normal blood vessel and non-calcified plaque, the shock waves produced by IVL do not modify the normal vessel tissue or non-calcified plaque. Moreover, IVL does not carry the same degree of risk of perforation, dissection, or other damage to vasculature as atherectomy procedures or angioplasty procedures using cutting or scoring balloons.

[0006] More specifically, catheters to deliver IVL therapy have been developed that include pairs of electrodes for electrohydraulically generating shock waves inside an angioplasty balloon. Shock wave devices can be particularly effective for treating calcified plaque lesions because the acoustic pressure from the shock waves can crack and disrupt lesions near the angioplasty balloon without harming the surrounding tissue. In these devices, the catheter is advanced over a guidewire through a patient's vasculature until it is positioned proximal to and/or aligned with a calcified plaque lesion in a body lumen. The balloon is then inflated with conductive fluid (using a relatively low pressure of 2-4 atm) so that the balloon expands to contact the lesion but is not an inflation pressure that substantively displaces the lesion. Voltage pulses can then be applied across the electrodes of the electrode pairs to produce acoustic shock waves that propagate through the walls of the angioplasty balloon and into the lesions. Once the lesions have been cracked by the acoustic shock waves, the balloon can be expanded further to increase the cross-sectional area of the lumen and improve blood flow through the lumen. Alternative devices to deliver IVL therapy can be within a closed volume other than an angioplasty balloon, such as a cap, balloons of variable compliancy, or other enclosure.

[0007] To date, commercially available IVL catheters have included shock wave emitters housed within an angioplasty balloon. While effective for treating calcified lesions, improvements to IVL catheters may be necessary to target other tissue types (e.g., fibrotic or thrombotic masses) or lesions with multiple tissue types. Further, for treating lesions other than stenotic lesions (e.g., blood clots, fibrotic tissue within anastomosis, or cancer tumors), employing an angioplasty balloon or another kind of enclosure may not be necessary.

SUMMARY

[0008] Described herein are devices and methods for generating shock waves in a body lumen. In some embodiments, a shock wave catheter includes, at its distal region, a retractable enclosure configured to enclose a shock wave emitter assembly in a closed configuration or to expose the shock wave emitter assembly to the vasculature (e.g., bloodstream) in an open configuration. In some embodiments, the catheter includes perfusion capabilities for circulating fluid at the treatment site.

[0009] According to an aspect of the disclosure, a catheter for emitting shock waves in a body lumen is provided, the catheter comprising: a distal member; a slidable elongate member having a proximal end and a distal end; an inner elongate member located at least partially inside of the slidable elongate member and extending to the distal member; a shock wave emitter assembly located on the inner elongate member and in a more distal location than the distal end of the slidable elongate member; and a retractable enclosure having a distal end, where: in a closed configuration, the distal end of the retractable enclosure extends to the distal member of the catheter, and in an open configuration, the distal end of the retractable enclosure is retracted from the distal member of the catheter.

[0010] In some aspects, the catheter further comprises a proximal handle having a switch that is operably coupled with the retractable enclosure to move the retractable membrane between the closed configuration and the open configuration. In some aspects, in the closed configuration, the distal end of the retractable enclosure and the distal member of the catheter form a seal. In some aspects, the distal member comprises an expandable member, the distal end of the retractable enclosure comprises a polymeric liner, and, in the closed configuration, the seal is formed between the expandable member and the polymeric liner. In some aspects, the polymeric liner comprises a polytetrafluoroethylene. In some aspects, in the closed configuration, the catheter further comprises a chamber surrounding the shock wave emitter assembly and having an internal pressure no less than 1 atm. In some aspects, the distal member comprises an expandable member located distally of the retractable enclosure. In some aspects, the retractable enclosure comprises at least one of polyether block amide, nylon, thermoplastic polyurethane, fluorinated ethylene propylene, and polytetrafluoroethylene. In some aspects, the distal member comprises an elastic tip. In some aspects, the catheter further comprises at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly. In some aspects, in the open configuration, the distal end of the retractable enclosure is spaced apart from the distal member of the catheter by a distance greater than or equal to 5 mm. In some aspects, in the open configuration, the distal end of the retractable enclosure is at least partially located proximally of the shock wave emitter assembly.

[0011] According to an aspect of the disclosure, a catheter for emitting shock waves in a body lumen is provided, the catheter comprising: a distal portion; an elongate member extending to the distal portion; a shock wave emitter assembly located on the elongate member; and a retractable enclosure having a distal end opening, where: in a closed configuration, the distal end opening of

the retractable enclosure extends to the distal portion, and in an open configuration, the distal end opening of the retractable enclosure is retracted from the distal portion.
[0012] In some aspects, the retractable enclosure comprises an angioplasty balloon. In some aspects, the distal end opening of the retractable enclosure includes an inner diameter no less than a maximum outer diameter of the shock wave emitter assembly. In some aspects, the distal end opening of the retractable enclosure includes an inner diameter no less than 5 mm. In some aspects, the catheter further comprises a proximal handle having a switch that is operably coupled with the retractable enclosure to move the retractable enclosure between the closed configuration and the open configuration. In some aspects, in the closed configuration, the distal end opening of the retractable enclosure and the distal member of the catheter form a seal. In some aspects, the distal member comprises an expandable member, the distal end opening of the retractable enclosure comprises a polymeric liner, and, in the closed configuration, the expandable member and the polymeric liner form the seal. In some aspects, the polymeric liner comprises polytetrafluoroethylene. In some aspects, in the closed configuration, the retractable enclosure surrounds the shock wave emitter assembly and has an internal pressure no less than 1 atm and no more than 4 atm. In some aspects, the retractable enclosure comprises at least one of polyether block amide and polytetrafluoroethylene. In some aspects, the distal member comprises an elastic tip. In some aspects, the catheter further comprises at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly. In some aspects, in the open configuration, the distal end opening of the retractable enclosure is spaced apart from the distal member of the catheter by a distance greater than or equal to 5 mm. In some aspects, in the open configuration, the distal end opening of the retractable enclosure is at least partially located proximally of the shock wave emitter assembly.
[0013] According to an aspect of the disclosure, a catheter for treating a lesion in a body lumen is provided, the catheter comprising: a distal member; an elongate member; a shock wave emitter assembly located at a distal region of the clongate member; a retractable enclosure having a distal end and configured to be retractably coupled to the distal member; and a proximal handle having a switch and located at a proximal end of the elongate member, where: when the switch is in a closed setting, the retractable enclosure is scaled to the distal member, and, when the switch is in an open setting, the retractable enclosure is spaced from the distal member.
[0014] In some aspects, the clongate member comprises a first lumen and the proximal handle comprises a first port in fluid communication with the first lumen. In some aspects, the clongate member further comprises a second lumen and the proximal handle further comprises a second port in fluid communication with the second lumen. In some aspects, the proximal handle further comprises a second port in fluid communication with the first lumen.
[0015] According to an aspect of the disclosure, a method of generating shock waves in a body lumen is provided, the method comprising: advancing a shock wave catheter through a body lumen, the shock wave catheter having: a distal member; an elongate member; a shock wave emitter assembly located at a distal region of the elongate member; and a retractable enclosure having a distal end and configured to be retractably coupled to the distal member; positioning the shock wave emitter assembly adjacent a treatment site; retracting the retractable enclosure; and generating one or more shock waves at the shock wave emitter assembly.
[0016] In some aspects, the method further comprises closing the retractable enclosure such that the distal end of the retractable enclosure is sealed to the distal member. In some aspects, an inner surface of the retractable enclosure comprises a polymeric liner that forms a fluid-tight seal with the distal member when the retractable enclosure and the distal member are coupled. In some aspects, the method further comprises depressurizing a chamber of the retractable enclosure such that a fluid-tight seal between the distal member and the retractable enclosure is broken. In some aspects, the method further comprises imaging the treatment site by one or more of x-ray fluoroscopy, intravascular ultrasound, and optical coherence tomography. In some aspects, the

method further comprises, based on imaging data, determining whether to generate shock waves with the retractable enclosure retracted or with the retractable enclosure sealed to the distal member.

[0017] According to an aspect of the disclosure, a method of generating shock waves is provided, the method comprising: advancing a shock wave catheter through a body lumen, the shock wave catheter having a shock wave emitter assembly; applying a first energy pulse to the shock wave emitter assembly to generate a first shock wave having a first sonic output; and after applying the first energy pulse, applying a second energy pulse having the same power as the first energy pulse to the shock wave emitter assembly to generate a second shock wave having a second sonic output that is greater than the first sonic output.

[0018] In some aspects, the shock wave catheter comprises a distal member, an elongate member, and a retractable enclosure having a distal end and configured to be retractably coupled to the distal member. In some aspects, the method further comprises, between applying the first energy pulse and applying the second energy pulse, retracting the retractable enclosure such that the retractable enclosure does not cover the shock wave emitter assembly.

[0019] According to an aspect of the disclosure, a catheter system for generating shock waves is provided, the catheter system comprising: a power source configured to generate a plurality of energy pulses, each energy pulse having a predetermined power; and a catheter comprising a shock wave emitter assembly and an elongate energy guide extending from the power source to the shock wave emitter assembly, where, in a first setting of the catheter system, the shock wave emitter assembly generates a first shock wave having a first sonic output when a first energy pulse of the plurality of energy pulses is applied to the shock wave emitter assembly, and, in a second setting of the catheter system, the shock wave emitter assembly generates a second shock wave having a second sonic output greater than the first sonic output when a second energy pulse of the plurality of energy pulses is applied to the shock wave emitter assembly.

[0020] In some aspects, the catheter further comprises a distal member and a retractable enclosure that, in the first setting, is sealed to the distal member so as to define a chamber and, in the second setting, is spaced from the distal member.

[0021] According to an aspect of the disclosure, a method of generating shock waves is provided, the method comprising: advancing a shock wave catheter through a body lumen, the shock wave catheter having a shock wave emitter assembly and a retractable enclosure; applying a first energy pulse to the shock wave emitter assembly to generate a first shock wave having a first sonic output; if the retractable enclosure is in a closed configuration, retracting the enclosure to expose the shock wave emitter assembly, or, if the retractable enclosure is in an open configuration, closing the retractable enclosure to enclose the shock wave emitter assembly; and applying a second energy pulse having a different power than the first energy pulse to the shock wave emitter assembly to generate a second shock wave.

Description

DESCRIPTION OF THE FIGURES

[0022] Illustrative aspects of the present disclosure are described in detail below with reference to the following figures. It is intended that the embodiments and figures disclosed herein are to be considered illustrative and exemplary rather than restrictive.

[0023] FIG. **1** illustrates a perspective view of an exemplary shock wave catheter having a retractable member, according to aspects of the disclosure.

[0024] FIG. **2**A illustrates a side view of an exemplary shock wave catheter having a retractable member in a sealed (closed) configuration, according to aspects of the disclosure.

[0025] FIG. **2**B illustrates a side view of the shock wave catheter of FIG. **2**A with the retractable

member in a retracted (open) configuration.
[0026] FIG. **3**A illustrates a perspective view of an exemplary shock wave catheter having a retractable enclosure in a sealed (closed) configuration, according to aspects of the disclosure.
[0027] FIG. **3**B illustrates a perspective view of the shock wave catheter of FIG. **3**A with the retractable enclosure in a retracted (open) configuration.
[0028] FIG. **4** illustrates a perspective view of an exemplary shock wave catheter having an inflatable distal member, according to aspects of the disclosure.
[0029] FIG. **5** illustrates a proximal region of an exemplary shock wave catheter, according to aspects of the disclosure.
[0030] FIG. **6** illustrates an exemplary method of generating shock waves, according to aspects of the disclosure.
[0031] FIG. **7** illustrates an exemplary method of generating shock waves, according to aspects of the disclosure.
[0032] FIG. **8** illustrates an exemplary method of generating shock waves, according to aspects of the disclosure.
[0033] FIG. **9** illustrates an exemplary computing system that may be included as part of a shock wave catheter system, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0034] The following description is presented to enable a person of ordinary skill in the art to make and use the various embodiments and aspects thereof disclosed herein. Descriptions of specific devices, assemblies, techniques, and applications are provided only as examples. Various modifications to the examples described herein will be readily apparent to those of ordinary skill in the art, and the general principles described herein may be applied to other examples and applications without departing from the spirit and scope of the various embodiments and aspects thereof. Thus, the various embodiments and aspects thereof are not intended to be limited to the examples described herein and shown but are to be accorded the scope consistent with the claims.
[0035] Efforts have been made to improve the design of electrode assemblies included in shock wave and directed cavitation catheters. For instance, low-profile electrode assemblies have been developed that reduce the crossing profile of a catheter and allow the catheter to more easily navigate calcified vessels to deliver shock waves in more severely occluded regions of vasculature. Examples of low-profile electrode designs can be found in U.S. Pat. Nos. 8,888,788, 9,433,428, and 10,709,462, and in U.S. Publication No. 2021/0085383, all of which are incorporated herein by reference in their entireties. Other catheter designs have improved the delivery of shock waves, for instance, by specific electrode construction and configuration thereby directing shock waves in a forward direction to break up tighter and harder-to-cross occlusions in vasculature. Examples of forward-firing catheter designs can be found in U.S. Pat. Nos. 10,966,737, 11,478,261, and 11,596,423, and in U.S. Publication Nos. 2023/0107690 and 2023/0165598, all of which are incorporated herein by reference in their entireties.
[0036] As used herein, the term “electrode” refers to an electrically conducting element (typically made of metal) that receives electrical current and subsequently releases the electrical current to another electrically conducting element. In the context of the present disclosure, electrodes are often positioned relative to each other, such as in an arrangement of an inner electrode and an outer electrode. Accordingly, as used herein, the term “electrode pair” refers to two electrodes that are positioned adjacent to each other such that application of a sufficiently high voltage to the electrode pair will cause an electrical current to transmit across the gap (also referred to as a “spark gap”) between the two electrodes (e.g., from an inner electrode to an outer electrode, or vice versa, optionally with the electricity passing through a conductive fluid or gas therebetween). In some contexts, one or more electrode pairs may also be referred to as an electrode assembly. In the context of the present disclosure, the term “emitter” broadly refers to the region of an electrode assembly where the current transmits across the electrode pair, generating a shock wave. The terms

"emitter sheath" and "emitter band" refer to a continuous or discontinuous band of conductive material that may form one or more electrodes of one or more electrode pairs, thereby forming a location of one or more emitters.

[0037] FIG. **1** illustrates a distal region of a shock wave catheter **100** having a retractable enclosure **110**, according to one or more aspects of the disclosure. The retractable enclosure **110** may include a distal sealing region **114**. An inner surface of the distal sealing region **114** may include a polymeric liner. The polymeric liner may be made of a lubricious material such as polytetrafluoroethylene (PTFE). According to some embodiments, the catheter **100** has a distal member **120**, which includes an expandable member **122** that, when in a closed configuration, expands in response to an internal pressure increase within the retractable enclosure **110**. The expandable member **122** may be sized such that it snugly fits in the distal sealing region **114** in the closed configuration and then volumetrically expands when the pressure within the retractable enclosure **110** is increased. When the expandable member **122** is in an expanded state as illustrated in FIG. **1**, a fluid-tight seal is maintained between the distal member **120** and the distal sealing region **114** of the retractable enclosure **110**. The expandable member **122** may be made of a compliant material. The expandable member **122** may be made of a low durometer material. In some embodiments, the distal member **120** includes an atraumatic, rounded, elastic tip.

[0038] In one or more embodiments, catheter **100** includes an elongate member **130** that extends from a proximal region (not shown in FIG. **1**) to the distal member **120**. In some embodiments, elongate member **130** includes one or more fluid lumens (e.g., to supply fluid for expanding the expandable member **122**). The elongate member **130** may include a guidewire lumen **132** that extends to the distal member **122** of the distal member **120**.

[0039] Catheter **100** includes a shock wave emitter assembly **140**, which is connected by one or more energy guides (e.g., electrical wires or optical fibers) to a power source (e.g., a high voltage power supply or a light energy source). In some embodiments, the shock wave emitter assembly **140** includes one or more emitters **141**, **142**, **143**, **144**. Each of the one or more emitters **141**, **142**, **143**, **144** may include one or more electrode pairs. When a sufficient voltage is applied across an electrode pair, a shock wave may be generated at the electrode pair. Each of these electrode pairs may be electrically connected (e.g., in series or in parallel) to one or more electrode pairs of one or more emitters. Accordingly, one high voltage pulse from a high voltage power source may generate multiple shock waves from catheter **100**. In some embodiments, one or more emitters are separately connected to a power source (e.g., by multiple channels) so that a single high voltage pulse at one channel only generates shock waves from emitters electrically connected to the one channel.

[0040] FIG. **2**A illustrates a closed configuration of shock wave catheter **100** where a distal end of retractable enclosure **110** extends to distal member **120**. In this configuration, retractable enclosure **110**, at least in part, defines a chamber **112** that houses the shock wave emitter assembly **140**. In some embodiments, chamber **112** is pressurized to a pressure of at least 1 atm in this closed configuration with a conductive fluid (e.g., saline). In some embodiments, chamber **112** is pressurized to a pressure of at most 5 atm. In some embodiments, chamber **112** is pressurized to a pressure that is no less than 2 atm and no greater than 4 atm. In some embodiments, retractable enclosure **110** is made of a material that does not expand when chamber **112** is pressurized to a pressure of up to 5 atm. Retractable enclosure **110** may extend distally from a distal end of a slidable elongate member **135**, which may extend to a proximal region of the catheter **100**. In some embodiments, as shown in FIGS. **2**A and **2**B, the slidable clongate member **135** may form an outer layer of the catheter **100**. In some embodiments, the slidable elongate member **135** be configured to slide longitudinally (in the distal-proximal direction) between two fixed concentric layers of the catheter **100**.

[0041] In some embodiments, in the closed configuration shown in FIG. **2**A, one or more shock waves are generated in the conductive fluid at emitters **141**, **142**, **143**, **144**. The one or more shock waves propagate through the fluid in chamber **112** and through retractable enclosure **110** and then reach a target (e.g., a calcified lesion, a thrombotic mass, or a fibrotic mass), where the one or more shock waves break up the target.

[0042] In contrast, FIG. **2**B illustrates an open configuration of catheter **100**, according to one or more embodiments. In the open configuration, retractable enclosure **110** is spaced from the distal member **230**. For example, the distal end of the retractable enclosure **110** may be spaced apart from the distal member **230** by a distance greater than or equal to 5 mm. In some embodiments, in the open configuration, a distal end of the retractable enclosure **110** is located more proximally than at least a part of shock wave emitter assembly **140**. In some embodiments, and as shown in FIG. **2**B, a distal end of the retractable enclosure **110** is located more proximally than emitters **141**, **142**, **143**, **144**.

[0043] In some embodiments, the effective sonic output of shock waves (i.e., the pressure exerted at the treatment site) may be higher in this open configuration than in the closed configuration. In some embodiments, shock wave emitter assembly **140**, when used in the open configuration, generates one or more cavitation bubbles, which expand to a size greater than that which would be possible within an enclosure. The collapse of a cavitation bubble in the open configuration generates additional pressure waves (e.g., from high-speed micro-jets) with greater energy than those generated from a cavitation bubble inside an enclosure. In some embodiments, depending on the acoustic properties of the enclosure, sonic output from a shock wave may be dampened upon propagation through an enclosure. As a result, catheter **100** may be employed in the open configuration when shock waves generated in the closed configuration do not sufficiently or efficiently treat target tissue.

[0044] In some embodiments, catheter **100** is connected (e.g., electrically or electromagnetically) to a power source. Depending on the intended use (e.g., in an open configuration or in a closed configuration), power supplied by the power source to catheter **100** may be adjusted or modulated to treat target tissue more effectively. In some embodiments, the power source may be a high voltage power supply and, depending on the intended use, properties of the supplied voltage pulse may be adjusted. For example, the amplitude of the voltage pulse may be increased or decreased when catheter **100** is switched from an open configuration to a closed configuration. In another example, the pulse width of a high voltage pulse may be adjusted to target a specific type of tissue (e.g., calcific, fibrotic, or thrombotic) or to be compatible with shock wave generation and propagation through a specific medium (e.g., saline or blood). In some embodiments, other pulse parameters (e.g., number of pulses, frequency, or number of IVL treatment cycles) may be optimized for a target tissue type and/or whether catheter **100** is intended for use in an open or closed configuration.

[0045] In some embodiments, retractable enclosure **110** is made of a tubular membrane. Retractable enclosure **110** may be made of a polyether block amide (e.g., Pebax), a nylon, a thermoplastic polyurethane (TPU), a fluorinated ethylene propylene (FEP), a polytetrafluoroethylene (PTFE), or a low durometer resin. Retractable enclosure **110** may have an outer diameter that is less than 15 mm, allowing catheter **100** to navigate through narrower and/or more tortuous body lumens than would be possible with conventional IVL devices. In some embodiments, retractable enclosure **110** has a length (i.e., in the catheter's distal-proximal direction) that is no less than a length of shock wave emitter assembly **140** (e.g., a distance from proximal end of emitter **141** to distal end of emitter **144**. In some embodiments, retractable enclosure **110** has a length between 3 mm and 100 mm. In some embodiments, retractable enclosure **110** has a length between 10 mm and 50 mm.

[0046] FIGS. **3**A and **3**B illustrate a distal region of a shock wave catheter **200**, according to one or more embodiments of the disclosure. Catheter **200** includes a retractable enclosure **210** that is retractable from a closed configuration (illustrated in FIG. **3**A) to an open configuration (illustrated in FIG. **3**B). The retractable enclosure **210** is attached to a distal region of a slidable elongate member **235**, which extends to a proximal end of the catheter **200**. In the closed configuration, retractable enclosure **210** surrounds and houses a shock wave emitter assembly **240** that is located on an elongate member **220**. In some embodiments, retractable enclosure **210** includes a central region **212** that expands in diameter when inflated with a fluid. In some embodiments, retractable enclosure **210** is an angioplasty balloon. In some embodiments, retractable enclosure **210** includes a distal opening **216** having an inner diameter that is no less than a maximum profile of shock wave emitter assembly **240**. In some embodiments, the inner diameter of the distal opening is no less than 5 mm.

[0047] In some embodiments, the retractable enclosure **210** includes, at an inner surface of its distal end, a polymeric liner **214** that, when coupled to a distal member **230** in the closed configuration, forms a fluid-tight seal. The fluidic seal may be formed by filling and pressuring the retractable enclosure **210** (e.g., to a pressure greater than 1 atm) such that an expandable member (e.g., an expandable dome or an expandable cone) of the distal member **230** seals with the polymeric liner **214**. The polymeric liner **214**, may be made of a compliant, low-durometer material.

[0048] In some embodiments, the retractable enclosure **210** may be used to exert outward pressure at the treatment site (e.g., along walls of a body lumen) after generation of shock waves. For example, shock waves may be generated at shock wave emitter assembly **240** with retractable enclosure **210** in a closed configuration. These shock waves may be generated with retractable enclosure **210** inflated to no more than 5 atm with a conductive fluid (e.g., saline). After shock wave generation, retractable enclosure **210** may be inflated to a pressure greater than 5 atm to achieve lumen dilation.

[0049] In another example, shock waves may be generated at shock wave emitter assembly **240** with retractable enclosure **210** in an open configuration. After shock wave generation, retractable enclosure **210** may be moved distally to the closed configuration. Upon scaling of retractable enclosure **210** to distal member **230**, retractable enclosure **210** may be inflated (e.g., to a pressure greater than 5 atm) to achieve lumen dilation.

[0050] FIG. **4** illustrates a distal region of another exemplary shock wave catheter **300**, according to one or more embodiments. Similar to catheters **100** and **200** described above, catheter **300** includes a retractable enclosure **310** that may be moved to generate shock waves from shock wave emitter assembly **340** in either a closed configuration or an open configuration. Catheter **300** additionally includes an occlusion member **350** located distally of shock wave emitter assembly **340**. Occlusion member **350** may be an expandable enclosure (e.g., an expandable balloon). In some embodiments, occlusion member **350** is made of a compliant, low durometer material. Occlusion member **350** may be made, for example, from a polyether block amide or a nylon material. Occlusion member **350** may be in a deflated or collapsed state during navigation through a body lumen and delivery to a treatment site. After positioning the shock wave emitter assembly **340** at the treatment site, occlusion member **350** may be expanded to occlude the body lumen and prevent flow of any debris generated during treatment to other parts of the patient's body. Occlusion member **350** may have a collapsed diameter and an inflated diameter greater than the collapsed diameter. In some embodiments, occlusion member **350** may be inflated by filling the member with a fluid via a fluid lumen, which may be separate from a fluid lumen for filling (and/or inflating) retractable enclosure **310**. In some embodiments, an IVL catheter may include a distal occlusion member (e.g., occlusion member **350**) and a proximal occlusion member, located proximally of a shock wave emitter assembly. In some embodiments, the use of one or more occlusion members to block debris flow may only be necessary when the IVL catheter is in an open configuration.

[0051] In some embodiments, an IVL catheter (e.g., catheter **100**, **200**, or **300**) includes a fluid supply lumen and a return lumen for flushing a treatment site with a fluid (e.g., saline) during IVL therapy. In some embodiments, fluid supply and return to the distal region of the catheter may be performed with a single lumen. In other words, a lumen that is used to fill or inflate a retractable enclosure may be used to evacuate any debris generated during shock wave treatment. Such infusion, aspiration, and suction features may be necessary particularly when the catheter is used in an open configuration when an enclosure is not positioned between the treatment site and the shock wave emitter assembly. In such examples, debris may be generated at the treatment site in response to a higher sonic pressure (relative to a closed configuration). Infusion of a conductive fluid via the supply line may ensure consistent sparking at one or more electrode pairs of the emitter assembly. In some embodiments, the infusion and/or aspiration rates at the treatment site are 5 mL/min to 25 mL/min.

[0052] Supply and/or return lumens may extend from a distal region of an IVL catheter to a proximal region of the IVL catheter. FIG. **5** illustrates an exemplary proximal handle **500** located in a proximal region of an IVL catheter, according to one or more embodiments. Proximal handle **500** may include an infusion port **510** that is in fluid communication with a supply lumen of the catheter and an aspiration port **520** that is in fluid communication with a return lumen of the catheter. One or both of the infusion port **510** and the aspiration port **520** may be fluidically connected to one or more pumps (e.g., syringe pumps or peristaltic pumps). The infusion port **510** may be in fluid communication with a fluid source for supplying conductive fluid (e.g., saline) to the treatment site (e.g., to fill and/or inflate an enclosure in a closed configuration or to flow fresh fluid to emitters in an open configuration). The proximal handle includes a power connector **540** through which a power source (not shown) may be connected. In some embodiments, the power source may be a high voltage generator. In some embodiments, the power source may be a light source (e.g., a laser).

[0053] In some embodiments, the proximal handle **500** includes a switch **530** that is operably connected to a retractable enclosure, such as retractable enclosure **210** of catheter **200** of FIG. **3**B, to move the retractable enclosure between a closed configuration to an open configuration. For example, a retractable enclosure may be connected to or integrally formed with an elongate member that extends to the switch **530**. Such an elongate member may be slidably located externally to an inner clongate member, such as externally to elongate member **220** of catheter **200** of FIG. **3**B.

[0054] In some embodiments, switch **530** may be movable (e.g., manually) to two or more stops corresponding to different positions of the retractable enclosure. In some embodiments, switch **530** may be slidable between two or more stops corresponding to different positions of the retractable enclosure. In some embodiments, switch **530** is movable between a first stop corresponding to a closed configuration of the retractable enclosure and a second stop corresponding to an open configuration of the retractable enclosure. In some embodiments, switch stops may correspond to a fully closed position, a fully retracted position, and one or more partially retracted positions of the retractable enclosure. For example, in embodiments of a shock wave emitter assembly having a plurality of emitters, different stop positions for the switch may correspond to different numbers of emitters being surrounded by the retractable enclosure.

[0055] FIG. **6** is a flowchart illustrating a method of using an IVL catheter with a retractable enclosure, according to some embodiments. At step **601**, the IVL catheter is advanced through a body lumen (e.g., a blood vessel). The retractable enclosure may be a tubular membrane, an angioplasty balloon, or another type of enclosure. During delivery in some embodiments, the enclosure is in a closed configuration. In some embodiments, the enclosure may be in a collapsed or deflated state to ensure a smaller cross-sectional profile during navigation through potentially tortuous body lumens. In some embodiments, the enclosure is in an open configuration during delivery.

[0056] At step **602**, the catheter may be positioned such that a shock wave emitter assembly (enclosed by the retractable enclosure in the closed configuration) of the catheter is proximate to a target treatment site. In some embodiments, the treatment site is imaged by one or more of x-ray fluoroscopy, intravascular ultrasound, or optical coherent tomography, and the catheter is positioned based on imaging data.

[0057] In some embodiments, at step **603**, the user (e.g., a physician) or an automated algorithm may determine whether the enclosure should be in a closed configuration or an open configuration depending on factors such as type of tissue being targeted, thickness of target lesion, eccentricity of target lesion, distance from target lesion, or other factors. In the open configuration, the effective sonic output may be substantially greater than in the closed configuration.

[0058] If the catheter is to be used with the enclosure in a closed configuration, at step **604**, the enclosure may be moved to a closed configuration if it is in an open configuration. In the closed configuration, a seal may be formed between a distal end of the enclosure and a distal member of the catheter. The distal member may include an expandable feature that may be expanded (e.g., by filling and pressuring the enclosure with a fluid) to create a seal that can withstand an internal pressure greater than 1 atm. The enclosure may be moved by the switch at the proximal handle of the catheter. The enclosure may be moved by a switch at a proximal handle of the catheter.

[0059] At step **605**, the chamber is filled with a fluid (e.g., saline) via a fluid lumen of the catheter. In some embodiments, the chamber is pressurized to a pressure no greater than 5 atm with the fluid. In some embodiments, the chamber is pressurized to a pressure that is no less than 2 atm and no greater than 4 atm.

[0060] At step **606**, shock waves are generated in the enclosure from the shock wave emitter assembly. In some embodiments, shock waves are generated by applying a high voltage across one or more electrode pairs of the shock wave emitter assembly. In other embodiments, shock waves are generated by light energy directed to the shock wave emitter assembly (e.g., by an optical fiber).

[0061] At step **607**, in some embodiments, the enclosure is optionally inflated to a pressure greater than 4 atm to achieve body lumen dilation.

[0062] In some embodiments, if the catheter is to be used with the enclosure in an open configuration, at step **614**, the enclosure may be retracted if it is in a closed configuration. The enclosure may be fully retracted to expose all of the shock wave emitter assembly, or partially retracted to partially expose the shock wave emitter assembly. The enclosure may be moved by the switch at the proximal handle of the catheter.

[0063] At step **615**, the target site may be infused with fluid (e.g., saline). In some embodiments, the site is also aspirated. The catheter may include separate supply and return lumens to enable infusion and aspiration. In some embodiments, a single lumen is used for both fluid supply and return. In some embodiments, before step **615**, the treatment site may be isolated within the vasculature by one or more expandable occlusion members (e.g., located distally and/or proximally of the treatment site) to ensure containment and aspiration of any debris generated during treatment.

[0064] At step **616**, one or more shock waves are generated at the shock wave emitter assembly. As noted above, shock waves generated in the open configuration may have an effective higher sonic output than in the closed configuration. That is, the acoustic pressure from shock waves generated in step **616** may be higher than the acoustic pressure from shock wave generated in step **606**.

[0065] After shock wave treatment, in some embodiments, the enclosure may optionally be moved to the closed configuration at step **617**. As in step **604**, a seal capable of withstanding an internal pressure greater than 1 atm may be formed in the closed configuration. Following closing the enclosure, in some embodiments, the enclosure is inflated to a pressure that is greater than 4 atm at step **607**.

[0066] After shock wave treatment at step **606** or **616**, optionally, step **603** may be repeated to switch the enclosure configuration and generate additional shock waves if necessary.

[0067] FIG. **7** is a flowchart illustrating a method of using an IVL catheter system including an IVL catheter configured to generate shock waves having different sonic outputs, according to some embodiments. At step **701**, the IVL catheter having a shock wave emitter assembly is advanced through a body lumen. At step **702**, the shock wave emitter assembly is positioned proximate to a treatment site (e.g., at a lesion of the body lumen).

[0068] In some embodiments, the body lumen is imaged (e.g., by x-ray fluoroscopy, optical coherence tomography, or intravascular ultrasound) to help position the catheter proximate to the lesion at step **702**. In some embodiments, the lesion is characterized (e.g., by an imaging method) to determine an appropriate therapy for the lesion (e.g., IVL catheter pulsing algorithm or IVL catheter configuration). For example, a preferred therapy to treat an eccentric lesion may be different from a preferred therapy to treat a concentric lesion. Further, different tissue types (e.g., calcific lesions or thrombotic lesions) having different acoustic properties may require different sonic treatment profiles to optimally treat a specific tissue type.

[0069] At step **703**, a first energy pulse is applied to the shock wave emitter assembly to generate a first shock wave having a first sonic output. The shock wave emitter assembly may include a plurality of shock wave emitters, where each of the shock wave emitters generates a separate shock wave.

[0070] At step **704**, a second energy pulse is applied to the shock wave emitter assembly to generate a second shock wave having a second sonic output higher than the first sonic output. The second energy pulse has the same power as the first energy pulse; for example, the first and second energy pulses may have the same voltage.

[0071] In some embodiments, between steps **703** and **704**, the IVL catheter system is changed from a first setting to a second setting. In some embodiments, in the first setting of the system, the IVL catheter is in a first configuration. In the second setting, the IVL catheter may be in a second configuration. In the first configuration, a retractable enclosure of the IVL catheter may be closed and, in the second configuration, the retractable enclosure may be open. In other embodiments, the IVL catheter system is changed from the first setting to the second setting by modifying electrical settings of the system.

[0072] FIG. **8** is a flowchart illustrating a method of using an IVL catheter system including an IVL catheter configured to generate shock waves having different sonic outputs, according to some embodiments. At step **801**, an IVL catheter having a shock wave emitter assembly and a retractable enclosure through a body lumen. At step **802**, the shock wave emitter assembly is positioned proximate to a treatment site (e.g., a lesion of the body lumen).

[0073] At step **803**, a first energy pulse is applied to the shock wave emitter assembly to generate a first shock wave having a first sonic output.

[0074] At step **804**, the IVL catheter is changed from a first configuration to a second configuration.

[0075] At step **805**, a second energy pulse having a different power than the first energy pulse is applied to the shock wave emitter assembly to generate a second shock wave. In some embodiments, the second shock wave has a sonic output equal to the first shock wave. In some embodiments, the second shock wave has a sonic output greater than the first shock wave. The power of the energy pulse may be changed at a power generator.

[0076] According to aspects of the disclosure, a method for manufacturing shock wave catheters may include: providing an inner elongate member; mounting one or more shock wave emitters at a distal end of the inner member; electrically connecting the one or more shock wave emitters to one or more wires; immobilizing the one or more wires to the inner member; providing a retractable member (e.g., a retractable enclosure) that is slidable along the inner member to surround the one or more shock wave emitters when the retractable member is in a distal-most position and to expose the one or more shock wave emitters when the retractable member is retracted from the distal-most position; and providing an outer member. The method may further include sterilizing (e.g., by electron beam sterilization or another form of radiation sterilization) components of the shock wave catheter.

[0077] According to aspects of the disclosure, a method of refurbishing a shock wave catheter may include replacing or repairing one or more components of the catheter, such as one or more of a retractable member, a distal member, an inner elongate member, and a slidable clongate member. For example, a retractable member may be replaced by detaching the retractable member from a slidable clongate member and attaching a new clongate member for the slidable elongate member. This may include sliding the slidable elongate member at least partially off of the catheter. In some variations, the retractable member may be refurbished either in place or after having been removed from the catheter. Refurbishment of the retractable member may include, for example, replacing or repairing a distal sealing region of the retractable enclosure. Refurbishing of a catheter may include replacing one or more components of a shock wave emitter assembly. This can include replacing one or more wires or removing a portion of one or more wires and soldering a new wire to the remaining portion and/or replacing one or more emitter sheaths. Optionally, an entire electrode assembly is removed from the catheter, repaired, and reassembled to the catheter. Optionally, an entire electrode assembly may be removed and replaced. Optionally, refurbishing a shock wave emitter assembly may include testing one or more performance parameters of a refurbished shock wave emitter. Testing may include testing the ability of the retractable member to seal to the distal member of the catheter, such as by positioning the retractable member in a sealed position, applying a pressure to the retractable member, and measuring the pressure to determine whether the applied pressure is above a predetermined threshold pressure. Testing may include testing the shock wave emitter assembly, such as by applying one or more voltage pulses to the shock wave emitter assembly and observing whether sparks are formed and/or measuring an intensity of the resulting shock waves. Testing may include testing the integrity of the retractable member by, e.g., pressurizing the retractable member in a closed position and monitoring pressure loss.

[0078] FIG. **9** illustrates an exemplary computing system **900** that may be included as part of a shock wave catheter system, in accordance with some examples of the disclosure. System **900** can be a client or a server. As shown in FIG. **9**, system **900** can be any suitable type of processor-based system, such as a personal computer, workstation, server, handheld computing device (portable electronic device) such as a phone or tablet, or dedicated device. The system **900** can include, for example, one or more of input device **920**, output device **930**, one or more processors **910**, storage **940**, and communication device **960**. Input device **920** and output device **930** can generally correspond to those described above and can either be connectable to, or integrated with, the computer.

[0079] Input device **920** can be any suitable device that provides input, such as a touch screen, keyboard or keypad, mouse, gesture recognition component of a virtual/augmented reality system, or voice-recognition device. Output device **930** can be or include any suitable device that provides output, such as a display, touch screen, haptics device, virtual/augmented reality display, or speaker.

[0080] Storage **940** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory including a RAM, cache, hard drive, removable storage disk, or other non-transitory computer readable medium. Communication device **960** can include any suitable device capable of transmitting and receiving signals over a network, such as a network interface chip or device. The components of the system **900** can be connected in any suitable manner, such as via a physical bus or wirelessly.

[0081] Processor(s) **910** can be any suitable processor or combination of processors, including any of, or any combination of, a central processing unit (CPU), graphics processing unit (GPU), field programmable gate array (FPGA), programmable system on chip (PSOC), and application-specific integrated circuit (ASIC). Software **950**, which can be stored in storage **940** and executed by one or more processors **910**, can include, for example, the programming that embodies the functionality or portions of the functionality of the present disclosure (e.g., as embodied in the devices as described above). The processor(s) **910** may be configured to process data received from sensors (e.g., imaging sensors, pressure sensors, or temperature sensors). The processor(s) **910** may be configured to control movement of a retractable enclosure of a shock wave catheter based, at least in part, on data received from the sensors and/or input from a user.

[0082] Software **950** can also be stored and/or transported within any non-transitory computer-readable storage medium for use by, or in connection with, an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as storage **940**, that can contain or store programming for use by, or in connection with, an instruction execution system, apparatus, or device.

[0083] Software **950** can also be propagated within any transport medium for use by, or in connection with, an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a transport medium can be any medium that can communicate, propagate, or transport programming for use by, or in connection with, an instruction execution system, apparatus, or device. The transport computer readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.

[0084] System **900** may be connected to a network, which can be any suitable type of interconnected communication system. The network can implement any suitable communications protocol and can be secured by any suitable security protocol. The network can comprise network links of any suitable arrangement that can implement the transmission and reception of network signals, such as wireless network connections, T1 or T3 lines, cable networks, DSL, or telephone lines.

[0085] Although shock wave devices described herein generate shock waves based on high voltage applied to electrodes, it should be understood that a shock wave device additionally or alternatively may comprise a laser and optical fibers as a shock wave emitter system whereby the laser source delivers energy through an optical fiber and into a fluid to form shock waves and/or cavitation bubbles.

[0086] Although the electrode assemblies and catheter devices described herein have been discussed primarily in the context of treating coronary occlusions, such as lesions in vasculature, the electrode assemblies and catheters herein can be used for a variety of occlusions, such as occlusions in the peripheral vasculature (e.g., above-the-knee, below-the-knee, iliac, or carotid). For further examples, similar designs may be used for treating soft tissues, such as cancer and tumors (i.e., non-thermal ablation methods), blood clots, fibroids, cysts, organs, scar and fibrotic tissue removal, or other tissue destruction and removal. Electrode assembly and catheter designs could also be used for neurostimulation treatments, targeted drug delivery, treatments of tumors in body lumens (e.g., tumors in blood vessels, the esophagus, intestines, stomach, or vagina), wound treatment, non-surgical removal and destruction of tissue, or used in place of thermal treatments or cauterization for venous insufficiency and fallopian ligation (i.e., for permanent female contraception).

[0087] In one or more examples, the electrode assemblies and catheters described herein could also be used for tissue engineering methods, for instance, for mechanical tissue decellularization to create a bioactive scaffold in which new cells (e.g., exogenous or endogenous cells) can replace the old cells; introducing porosity to a site to improve cellular retention, cellular infiltration/migration, and diffusion of nutrients and signaling molecules to promote angiogenesis, cellular proliferation, and tissue regeneration similar to cell replacement therapy. Such tissue engineering methods may be useful for treating ischemic heart disease, fibrotic liver, fibrotic bowel, and traumatic spinal cord injury (SCI). For instance, for the treatment of SCI, the devices and assemblies described herein could facilitate the removal of scarred spinal cord tissue, which acts like a barrier for neuronal reconnection, before the injection of an anti-inflammatory hydrogel loaded with lentivirus to genetically engineer the spinal cord neurons to regenerate.

[0088] The elements and features of the exemplary electrode assemblies and catheters discussed above may be rearranged, recombined, and modified, without departing from the present invention. Furthermore, numerical designators such as “first”, “second”, “third”, “fourth”, etc. are merely descriptive and do not indicate a relative order, location, or identity of elements or features described by the designators. For instance, a “first” shock wave may be immediately succeeded by a “third” shock wave, which is then succeeded by a “second” shock wave. As another example, a “third” emitter may be used to generate a “first” shock wave and vice versa. Accordingly, numerical designators of various elements and features are not intended to limit the disclosure and may be modified and interchanged without departing from the subject invention.
[0089] It should be noted that the elements and features of the example catheters illustrated throughout this specification and drawings may be rearranged, recombined, and modified without departing from the present invention. For instance, while this specification and drawings describe and illustrate catheters having several example balloon designs, the present disclosure is intended to include catheters having a variety of balloon configurations. The number, placement, and spacing of the electrode pairs of the shock wave generators can be modified without departing from the subject invention. Further, the number, placement, and spacing of balloons of catheters can be modified without departing from the subject invention.
[0090] It should be understood that the foregoing is only illustrative of the principles of the invention, and that various modifications, alterations, and combinations can be made by those skilled in the art without departing from the scope and spirit of the invention. Any of the variations of the various catheters disclosed herein can include features of any other catheters or combination of catheters described herein. Furthermore, any of the methods can be used with any of the catheters disclosed. Accordingly, it is not intended that the invention be limited, except as by the appended claims.
EXEMPLARY EMBODIMENTS
[0091] The following embodiments are exemplary and are not intended to limit the scope of any invention described herein.
[0092] Embodiment 1. A catheter for emitting shock waves in a body lumen, the catheter comprising: [0093] a distal member; [0094] a slidable elongate member having a proximal end and a distal end; [0095] an inner elongate member located at least partially inside of the slidable elongate member and extending to the distal member; [0096] a shock wave emitter assembly located on the inner elongate member and in a more distal location than the distal end of the slidable elongate member; and [0097] a retractable enclosure having a distal end, [0098] where: [0099] in a closed configuration, the distal end of the retractable enclosure extends to the distal member of the catheter, and [0100] in an open configuration, the distal end of the retractable enclosure is retracted from the distal member of the catheter.
[0101] Embodiment 2. The catheter of embodiment 1, further comprising a proximal handle having a switch that is operably coupled with the retractable enclosure to move the retractable enclosure between the closed configuration and the open configuration.
[0102] Embodiment 3. The catheter of embodiment 2, wherein the slidable elongate member comprises a first lumen and the proximal handle comprises a first port in fluid communication with the first lumen.
[0103] Embodiment 4. The catheter of embodiment 3, wherein the slidable elongate member further comprises a second lumen and the proximal handle further comprises a second port in fluid communication with the second lumen.
[0104] Embodiment 5. The catheter of embodiment 3 or 4, wherein the proximal handle further comprises a second port in fluid communication with the first lumen.
[0105] Embodiment 6. The catheter of any one of embodiments 1-5, wherein, in the closed configuration, the distal end of the retractable enclosure and the distal member of the catheter form a seal.

[0106] Embodiment 7. The catheter of embodiment 6, wherein the distal member comprises an expandable member, the distal end of the retractable enclosure comprises a polymeric liner, and, in the closed configuration, the seal is formed between the expandable member and the polymeric liner.
[0107] Embodiment 8. The catheter of embodiment 7, wherein the polymeric liner comprises a polytetrafluoroethylene.
[0108] Embodiment 9. The catheter of any one of embodiments 1-8, wherein, in the closed configuration, the catheter further comprises a chamber surrounding the shock wave emitter assembly and having an internal pressure no less than 1 atm.
[0109] Embodiment 10. The catheter of any one of embodiments 1-9, wherein the distal member comprises an expandable member located distally of the retractable enclosure.
[0110] Embodiment 11. The catheter of any one of embodiments 1-10, wherein the retractable enclosure comprises at least one of polyether block amide, nylon, thermoplastic polyurethane, fluorinated ethylene propylene, and polytetrafluoroethylene.
[0111] Embodiment 12. The catheter of any one of embodiments 1-11, wherein the distal member comprises an elastic tip.
[0112] Embodiment 13. The catheter of any one of embodiments 1-12, further comprising at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly.
[0113] Embodiment 14. The catheter of any one of embodiments 1-13, wherein, in the open configuration, the distal end of the retractable enclosure is spaced apart from the distal member of the catheter by a distance greater than or equal to 5 mm.
[0114] Embodiment 15. The catheter of any one of embodiments 1-14, wherein, in the open configuration, the distal end of the retractable enclosure is at least partially located proximally of the shock wave emitter assembly.
[0115] Embodiment 16. The catheter of any one of embodiments 1-15, wherein the retractable enclosure comprises an angioplasty balloon.
[0116] Embodiment 17. The catheter of embodiment 16, wherein the retractable enclosure includes a distal end opening with an inner diameter no less than a maximum outer diameter of the shock wave emitter assembly.
[0117] Embodiment 18. The catheter of embodiment 17, wherein the inner diameter of the distal end opening of the retractable enclosure is no less than 5 mm.
[0118] Embodiment 19. The catheter of any one of embodiments 16-18, wherein, in the closed configuration, the retractable enclosure surrounds the shock wave emitter assembly and has an internal pressure no less than 1 atm and no more than 4 atm.
[0119] Embodiment 20. The catheter of any one of embodiments 16-19, wherein the retractable enclosure comprises at least one of polyether block amide and polytetrafluoroethylene.
[0120] Embodiment 21. The catheter of any one of embodiments 16-20, further comprising at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly.
[0121] Embodiment 22. The catheter of any one of embodiments 16-21, wherein, in the open configuration, a distal end opening of the retractable enclosure is spaced apart from the distal member of the catheter by a distance greater than or equal to 5 mm.
[0122] Embodiment 23. A method of generating shock waves in a body lumen, the method comprising: [0123] advancing a shock wave catheter through the body lumen, the shock wave catheter having: [0124] a distal member; [0125] an elongate member; [0126] a shock wave emitter assembly located at a distal region of the elongate member; and [0127] a retractable enclosure having a distal end and configured to be retractably coupled to the distal member; [0128] positioning the shock wave emitter assembly adjacent a treatment site; [0129] retracting the retractable enclosure; and [0130] generating one or more shock waves at the shock wave emitter assembly.
[0131] Embodiment 24. The method of embodiment 23, further comprising closing the retractable

enclosure such that the distal end of the retractable enclosure is sealed to the distal member.
[0132] Embodiment 25. The method of embodiment 23 or 24, further comprising depressurizing a chamber of the retractable enclosure such that a fluid-tight seal between the distal member and the retractable enclosure is broken.
[0133] Embodiment 26. The method of any one of embodiments 23-25, further comprising imaging the treatment site by one or more of x-ray fluoroscopy, intravascular ultrasound, and optical coherence tomography.
[0134] Embodiment 27. The method of embodiment 26, further comprising, based on imaging data, determining whether to generate shock waves with the retractable enclosure retracted or with the retractable enclosure sealed to the distal member.
[0135] Embodiment 28. A method of generating shock waves, the method comprising: [0136] advancing a shock wave catheter through a body lumen, the shock wave catheter having a shock wave emitter assembly; [0137] applying a first energy pulse to the shock wave emitter assembly to generate a first shock wave having a first sonic output; and [0138] after applying the first energy pulse, applying a second energy pulse having the same power as the first energy pulse to the shock wave emitter assembly to generate a second shock wave having a second sonic output that is greater than the first sonic output.
[0139] Embodiment 29. The method of embodiment 28, wherein the shock wave catheter comprises a distal member, an elongate member, and a retractable enclosure having a distal end and configured to be retractably coupled to the distal member.
[0140] Embodiment 30. The method of embodiment 29, further comprising, between applying the first energy pulse and applying the second energy pulse, retracting the retractable enclosure such that the retractable enclosure does not cover the shock wave emitter assembly.
[0141] Embodiment 31. A catheter system for generating shock waves, the catheter system comprising: [0142] a power source configured to generate a plurality of energy pulses, each energy pulse having a predetermined power; and [0143] a catheter comprising a shock wave emitter assembly and an elongate energy guide extending from the power source to the shock wave emitter assembly, [0144] where, in a first setting of the catheter system, the shock wave emitter assembly generates a first shock wave having a first sonic output when a first energy pulse of the plurality of energy pulses is applied to the shock wave emitter assembly, and, in a second setting of the catheter system, the shock wave emitter assembly generates a second shock wave having a second sonic output greater than the first sonic output when a second energy pulse of the plurality of energy pulses is applied to the shock wave emitter assembly.
[0145] Embodiment 32. The catheter system of embodiment 31, wherein the catheter further comprises a distal member and a retractable enclosure that, in the first setting, is sealed to the distal member so as to define a chamber and, in the second setting, is spaced from the distal member.
[0146] Embodiment 33. A method of generating shock waves, the method comprising: [0147] advancing a shock wave catheter through a body lumen, the shock wave catheter having a shock wave emitter assembly and a retractable enclosure; [0148] applying a first energy pulse to the shock wave emitter assembly to generate a first shock wave having a first sonic output; [0149] if the retractable enclosure is in a closed configuration, retracting the retractable enclosure to expose the shock wave emitter assembly, or, if the retractable enclosure is in an open configuration, closing the retractable enclosure to enclose the shock wave emitter assembly; and applying a second energy pulse having a different power than the first energy pulse to the shock wave emitter assembly to generate a second shock wave.

Claims

1. A catheter for emitting shock waves in a body lumen, the catheter comprising: a distal member; a slidable elongate member having a proximal end and a distal end; an inner elongate member located at least partially inside of the slidable elongate member and extending to the distal member; a shock wave emitter assembly located on the inner elongate member and in a more distal location than the distal end of the slidable elongate member; and a retractable enclosure having a distal end, where: in a closed configuration, the distal end of the retractable enclosure extends to the distal member of the catheter to form a seal, and in an open configuration, the distal end of the retractable enclosure is retracted from the distal member of the catheter.

2. The catheter of claim 1, further comprising a proximal handle having a switch that is operably coupled with the retractable enclosure to move the retractable enclosure between the closed configuration and the open configuration.

3. The catheter of claim 2, wherein the slidable elongate member comprises a first lumen and the proximal handle comprises a first port in fluid communication with the first lumen.

4. The catheter of claim 3, wherein the slidable elongate member further comprises a second lumen and the proximal handle further comprises a second port in fluid communication with the second lumen.

5. The catheter of claim 3, wherein the proximal handle further comprises a second port in fluid communication with the first lumen.

6. (canceled)

7. The catheter of claim 1, wherein the distal member comprises an expandable member, the distal end of the retractable enclosure comprises a polymeric liner, and, in the closed configuration, the seal is formed between the expandable member and the polymeric liner.

8. The catheter of claim 7, wherein the polymeric liner comprises a polytetrafluoroethylene.

9. The catheter of claim 1, wherein, in the closed configuration, the catheter further comprises a chamber surrounding the shock wave emitter assembly and having an internal pressure no less than 1 atm.

10. The catheter of claim 1, wherein the distal member comprises an expandable member located distally of the retractable enclosure.

11. The catheter of claim 1, wherein the retractable enclosure comprises polyether block amide, nylon, thermoplastic polyurethane, fluorinated ethylene propylene, polytetrafluoroethylene, or a combination thereof.

12. The catheter of claim 1, wherein the distal member comprises an elastic tip.

13. The catheter of claim 1, further comprising at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly.

14. The catheter of claim 1, wherein, in the open configuration, the distal end of the retractable enclosure is spaced apart from the distal member of the catheter by a distance greater than or equal to 5 mm.

15. The catheter of claim 1, wherein, in the open configuration, the distal end of the retractable enclosure is at least partially located proximally of the shock wave emitter assembly.

16. The catheter of claim 1, wherein the retractable enclosure comprises an angioplasty balloon.

17. The catheter of claim 16, wherein the retractable enclosure includes a distal end opening with an inner diameter no less than a maximum outer diameter of the shock wave emitter assembly.

18. The catheter of claim 17, wherein the inner diameter of the distal end opening of the retractable enclosure is no less than 5 mm.

19. The catheter of claim 16, wherein, in the closed configuration, the retractable enclosure forms a chamber surrounding the shock wave emitter assembly, the chamber having an internal pressure no less than 1 atm and no more than 4 atm.

20. The catheter of claim 16, wherein the retractable enclosure comprises polyether block amide, polytetrafluoroethylene, or a combination thereof.

21. The catheter of claim 16, further comprising at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly.

22. The catheter of claim 16, wherein, in the open configuration, a distal end opening of the

retractable enclosure is spaced apart from the distal member of the catheter by a distance greater than or equal to 5 mm.
23-30. (canceled)
31. A catheter system for generating shock waves, the catheter system comprising: a power source configured to generate a plurality of energy pulses, each energy pulse having a predetermined power; and a catheter comprising a shock wave emitter assembly and an elongate energy guide extending from the power source to the shock wave emitter assembly, where, in a first setting of the catheter system, the shock wave emitter assembly generates a first shock wave having a first sonic output when a first energy pulse of the plurality of energy pulses is applied to the shock wave emitter assembly, and, in a second setting of the catheter system, the shock wave emitter assembly generates a second shock wave having a second sonic output greater than the first sonic output when a second energy pulse of the plurality of energy pulses is applied to the shock wave emitter assembly, and wherein the catheter further comprises a distal member and a retractable enclosure that, in the first setting, is sealed to the distal member so as to define a chamber and, in the second setting, is spaced from the distal member.
32. (canceled)
33. (canceled)
34. The system of claim 31, wherein the catheter comprises: a slidable elongate member; and an inner elongate member located at least partially inside of the slidable elongate member and extending to the distal member, wherein the shock wave emitter assembly is located on the inner elongate member and in a more distal location than a distal end of the slidable elongate member.
35. The system of claim 34, wherein the catheter further comprises a proximal handle having a first port, and the slidable elongate member comprises a first lumen in fluid communication with the first lumen.
36. The system of claim 35, wherein the proximal handle further comprises a second port, and the slidable elongate member further comprises a second lumen in fluid communication with the second port.
37. The system of claim 35, wherein the proximal handle further comprises a second port in fluid communication with the first lumen of the slidable elongate member.
38. The system of claim 31, wherein the catheter further comprises a proximal handle having a switch that is operably coupled with the retractable enclosure to move the retractable enclosure between the first setting and the second setting.
39. The system of claim 31, wherein the distal member comprises an expandable member, the distal end of the retractable enclosure comprises a polymeric liner, and, in the first setting, the expandable member and the polymeric liner are sealed.
40. The system of claim 31, wherein the distal member comprises an expandable member located distally of the retractable enclosure.
41. The system of claim 31, wherein the distal member comprises an elastic tip.
42. The system of claim 31, wherein, in the first setting, the retractable enclosure forms a chamber surrounding the shock wave emitter assembly, the chamber having an internal pressure no less than 1 atm and no more than 4 atm.
43. The system of claim 31, wherein the catheter further comprises at least one fluid lumen having a distal opening adjacent the shock wave emitter assembly.
44. The system of claim 31, wherein, in the second setting, a distal end of the retractable enclosure is spaced apart from the distal member of the catheter by a distance of greater than or equal to 5 mm.
45. The system of claim 31, wherein, in the second setting, a distal end of the retractable enclosure is at least partially located proximally of the shock wave emitter assembly.
46. The system of claim 31, wherein the retractable enclosure comprises a distal end opening having an inner diameter no less than a maximum outer diameter of the shock wave emitter

assembly.

47. The system of claim 46, wherein the inner diameter of the distal end opening of the retractable enclosure is no less than 5 mm.

48. The system of claim 31, wherein the power source is configured to adjust an amplitude, a pulse width, a frequency, a number of treatment cycles, a number of pulses generated by the shock wave emitter assembly, or a combination thereof between the first setting and the second setting.